Fault-Tolerant Multi-GPU LLM Serving with Adaptive KV-Cache Checkpointing

Progress Update

Scope

All results are from a **proof-of-concept setup**:

- 1B model, 2 GPUs, synthetic workloads
- Goal: validate the end-to-end pipeline

Production-scale evaluation (8B model, 4+ GPUs, real datasets) is the next step.

Problem

- Single-node multi-GPU LLM serving: replicas can fail (GPU reset, OOM, driver crash...)
- Failure -> capacity loss -> interrupted streams, SLO violations, goodput drop
- Naive checkpoint: too frequent = high overhead; too rare = long recovery
- **Core question: how to jointly optimize routing, admission, and adaptive checkpointing to maximize goodput while meeting SLOs under failures?**

Our System

(1) Benders-based Robust Scheduler

- Periodic decision epochs on system snapshot
- Master: admission + routing decisions
- Subproblem: verify recovery per failure scenario
- Pooled-flow screening + logic-based cuts

(2) Adaptive Checkpoint Policy

- Per-request, triggered on new stable KV blocks
- Publish when: Dreplay > Dload + lambda * Dckpt
- Early: skip (cheap to replay) / Late: checkpoint (expensive to lose)

Ablation Matrix

	Fixed Checkpoint	Adaptive Checkpoint
Greedy Routing	Fixed-Low / Fixed-High	Checkpoint-Only
Benders Routing	Robust-Routing-Only	Our-System

- Plus **No-FT** (no fault tolerance) as overhead-free reference

Experiment Setup

Hardware & Model

- Llama-3.2-1B-Instruct, 2 GPUs (DP replicas), 8GB checkpoint pool (host RAM)

Workloads

Workload	Prompt	Output	Arrival
W1 Short Interactive	50-200	20-100	Poisson
W2 Long Generation	200-500	200-500	Poisson
W3 Bursty Mixed	50-500	20-500	Bursty

- **Poisson**: steady rate, random intervals (stable load)
- **Bursty**: 15s normal -> 5s at 3x rate -> repeat (traffic spikes)

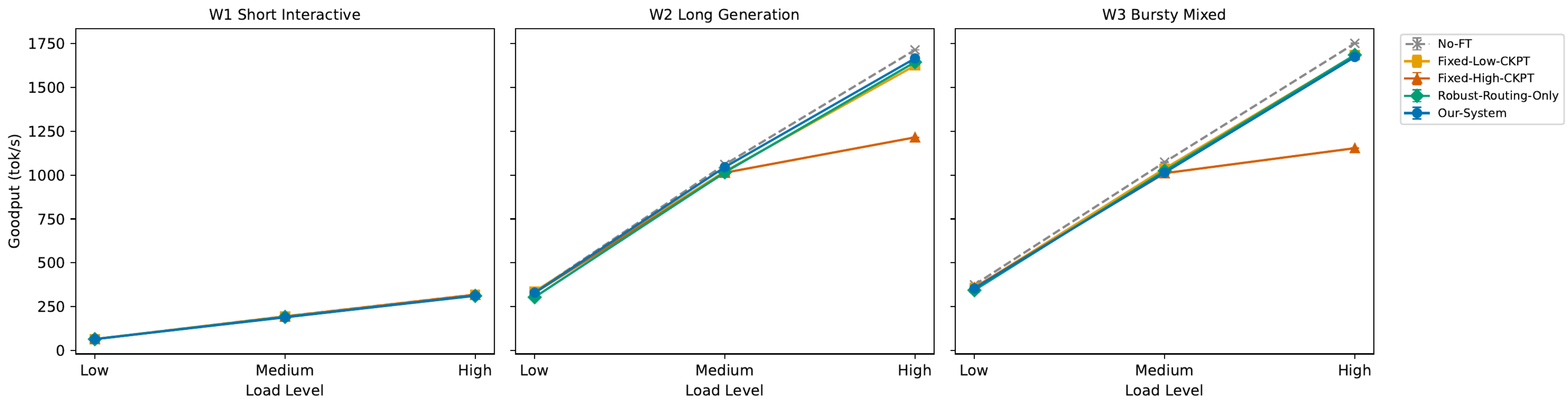
Load & Faults

- Load: Low (1), Medium (3), High (5 rps) / Faults: none / F1 (15s) / F2 (30s) / F3 (50s)
- Duration: 70s / SLO: TTFT 2s, TPOT 100ms, Failover gap 3s

5 Experiments

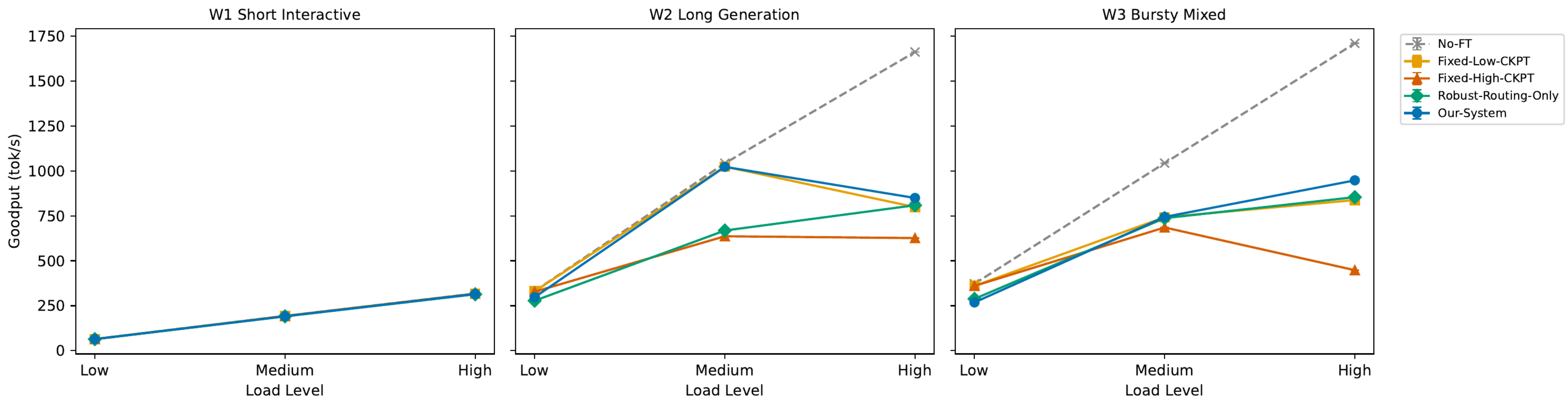
Experiment	Goal	Key Metrics
E1: Main	End-to-end comparison	Goodput, SLO violation, failover gap
E2: Recovery	Recovery time breakdown	Detection / KV Restore / Replay
E3: Ablation	Routing vs Checkpoint contrib.	Goodput by component
E4: Tradeoff	Ckpt overhead vs recovery	Goodput + failover gap
E5: Controller	Solver overhead	Solver latency per epoch

E1: Goodput -- No Fault



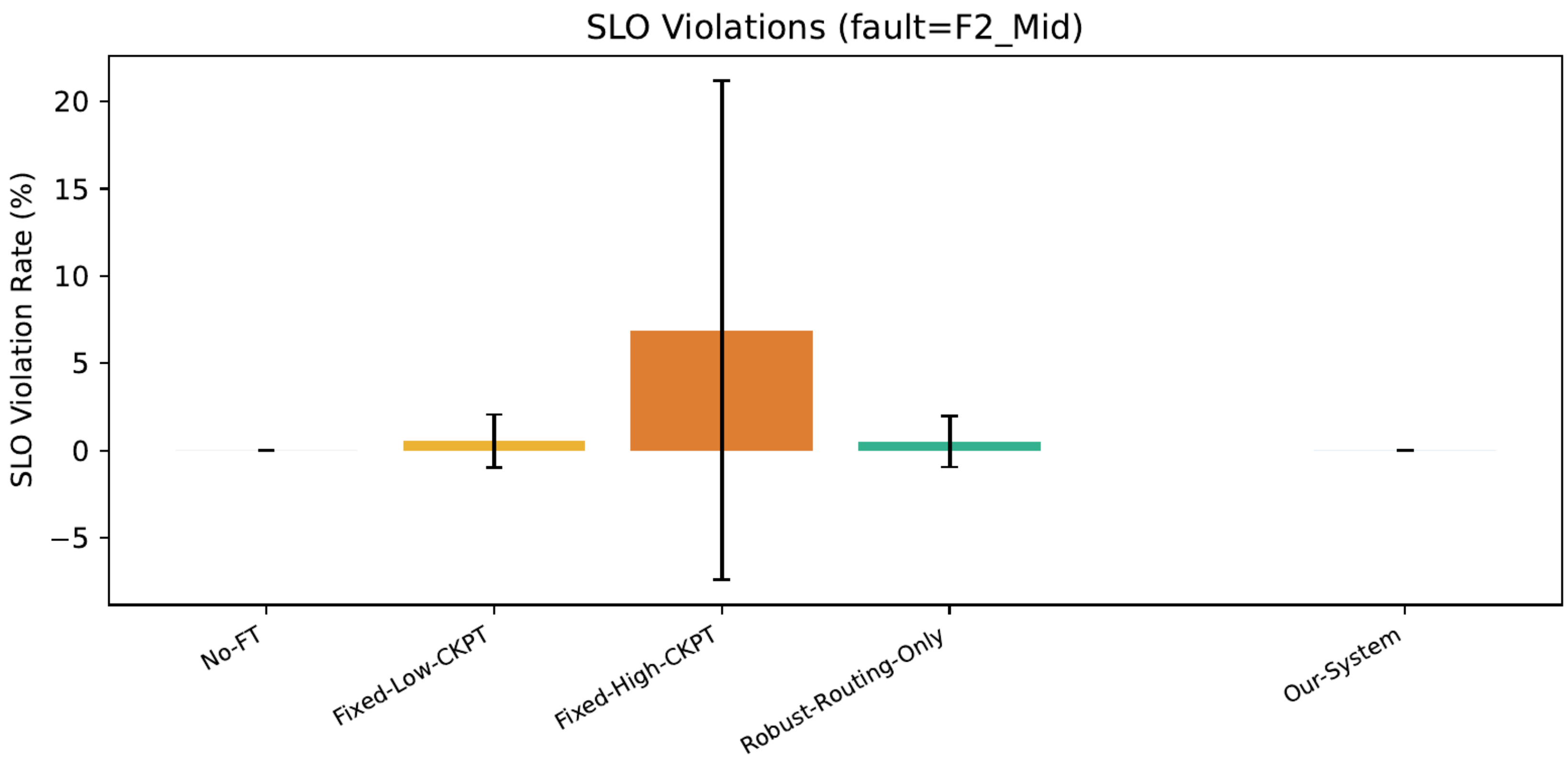
- All methods ~ No-FT -> **near-zero normal-case overhead**
- Exception: **Fixed-High-CKPT** drops ~29% on W2/High (1216 vs 1715 tok/s)

E1: Goodput -- With Fault (F2_Mid)



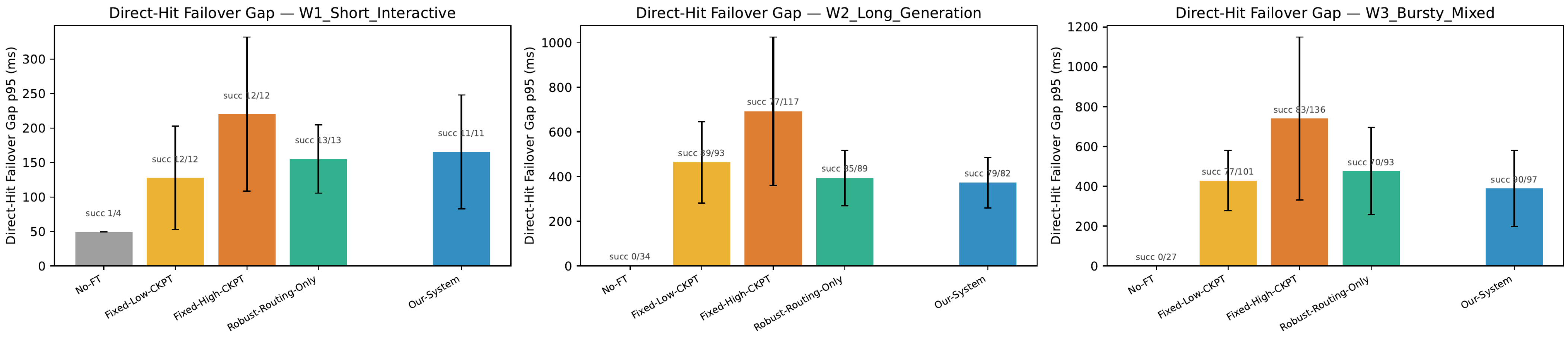
- No-FT **silently drops fault-hit requests** (completion 97-99%, 0% failover)
- Our-System matches Fixed-Low on W2/Med (~1040) with 100% completion
- Fixed-High-CKPT worst: overhead + slow recovery + low completion

E1: SLO Violations (F2_Mid)



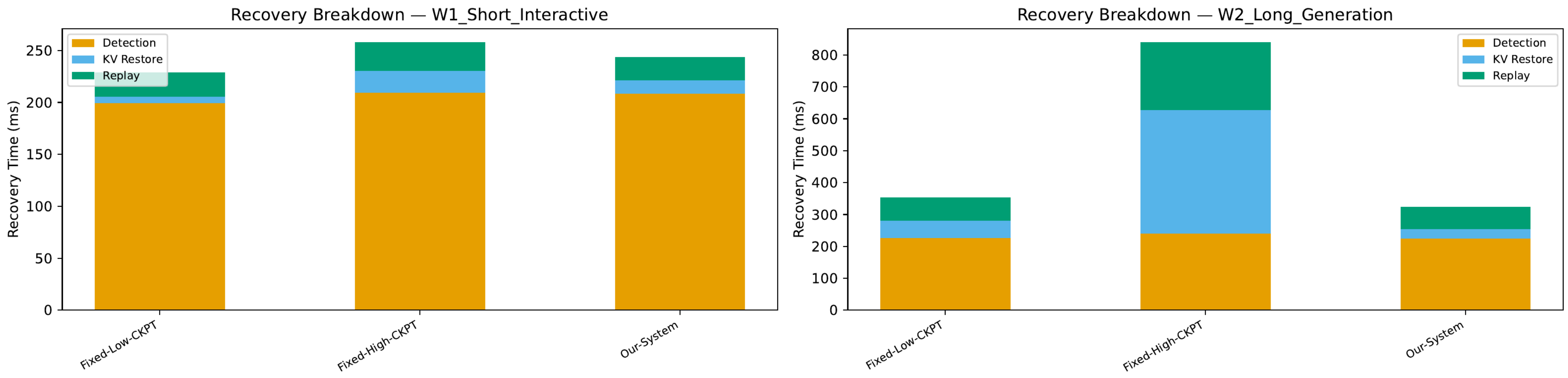
- With fault: **Fixed-High-CKPT** avg ~7% (worst 20%+); **Our-System ~ 0%**
- Without fault: all methods 0% violation (omitted -- no differentiation)

E1: Failover Gap (p95)



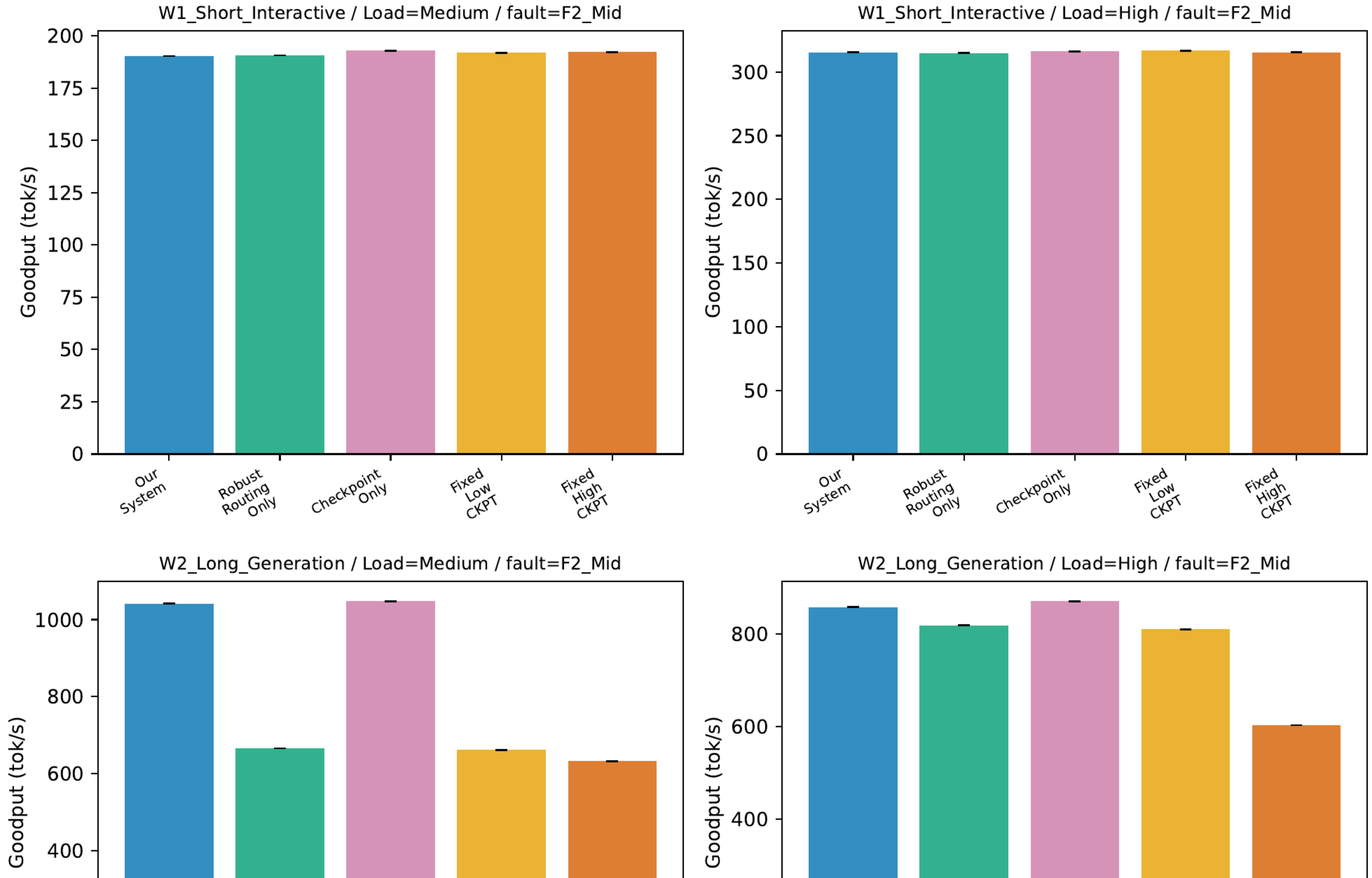
- No-FT: 0% recovery on W2/W3 (succ 0/34, 0/27)
- Fixed-High: highest gap (~700ms) + lowest success (77/117 = 66%)
- **Our-System: moderate gap + highest success rate** (W3: 90/97 = 93%)

E2: Recovery Time Breakdown

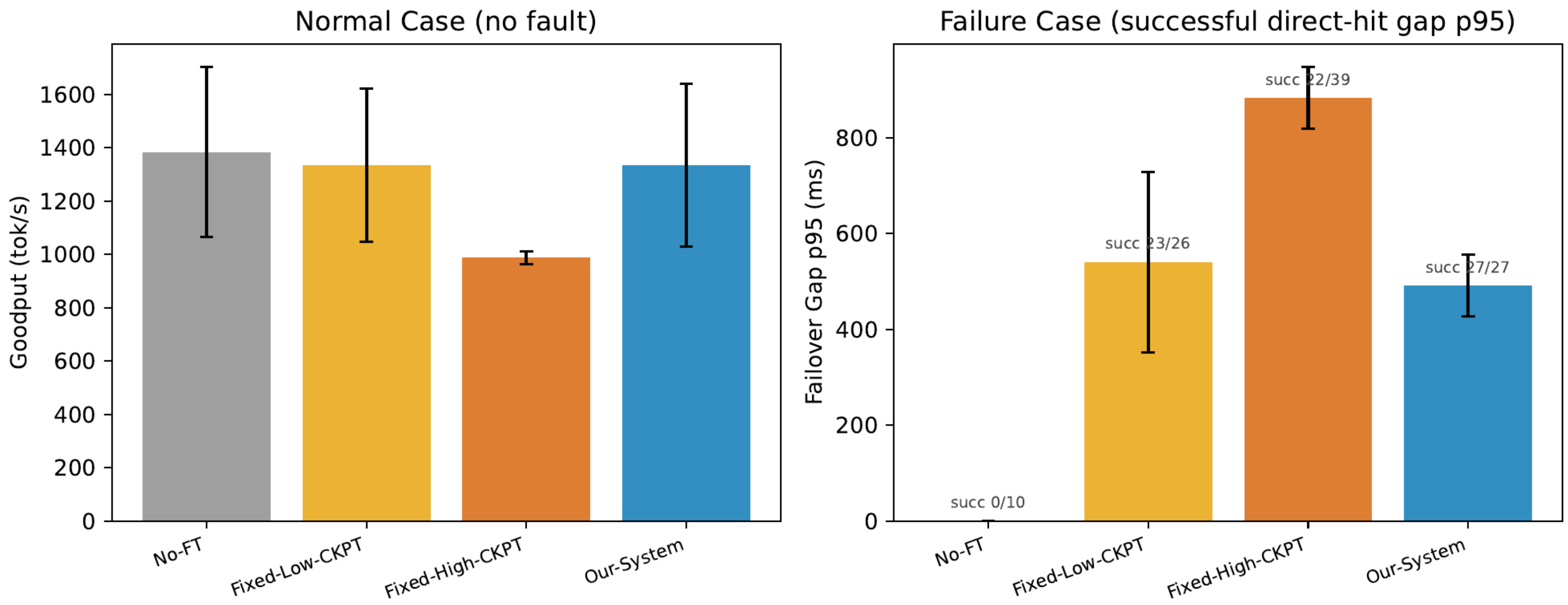


- W1 (short): detection dominates (~200ms), all similar
- W2 (long): Fixed-High ~830ms; Fixed-Low ~350ms; **Our-System ~330ms**
- Adaptive ckpt finds **sweet spot**: not too much to load, not too much to replay

E3: Ablation -- With Fault (F2_Mid)

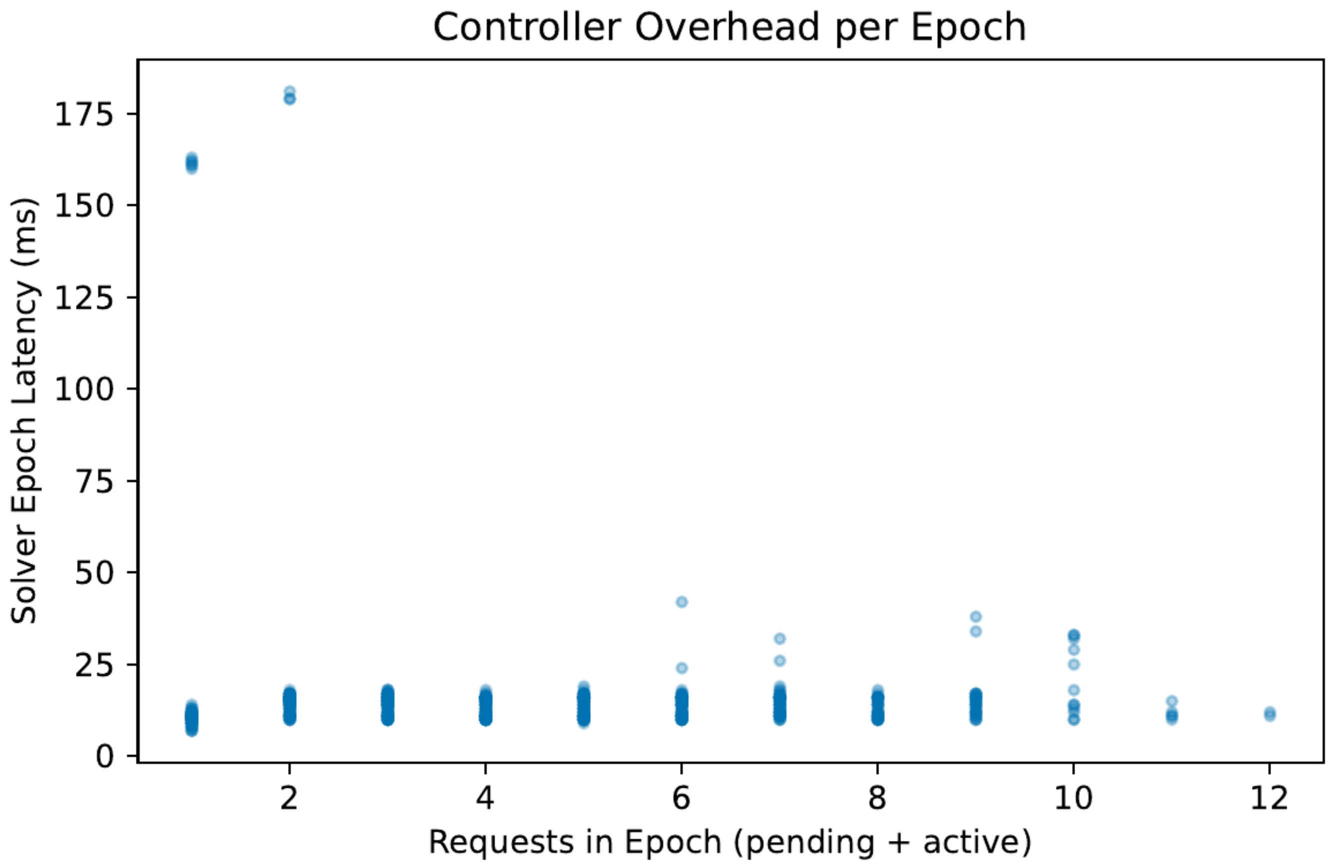


E4: Checkpoint Overhead vs Recovery Benefit



- Left (normal): Fixed-High loses ~29%; others ~ No-FT
- Right (fault): No-FT 0/10; Fixed-High 22/39 (56%), gap ~883ms
- **Our-System: 27/27 (100%), gap ~491ms -- best on both axes**
- **More checkpointing != better**

E5: Controller Overhead



- Median solver latency: **~15ms/epoch**, flat across 1-12 requests
- Outliers 160-180ms: cold start (first 1-2 calls), fixable via warmup
- Acceptable for online serving (decode step is ms-scale)

Summary

Key Findings

- 1. **Near-zero normal-case overhead**: goodput matches No-FT
- 2. **Best under failures**: highest recovery rate + lowest SLO violations
- 3. **More checkpointing != better**: Fixed-High hurts both goodput and recovery
- 4. **Adaptive checkpoint is the primary contributor**; routing matters more at scale
- 5. **Solver overhead is small**: ~15ms per epoch

Next Steps

- Real datasets: ShareGPT / CNN-DailyMail / Alpaca (real length distributions)
- Lightweight output-length predictor for better admission decisions
- Scale to 8B model on 4 GPUs (dp=4, losing 1 GPU = 25% capacity loss)
- Profile solver parameters from actual hardware measurements
- Per-request SLO differentiation (e.g. chat 100ms, summarization 200ms)
- 3 seeds, 300s runs for statistical significance